

AB POWER ENGINEERING

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ENERGY AND POWER

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1. Property that enable things to do work or cause change. ☒ a. Energy b. Power c. Fuel d. None of the above	4. Unit of power consumption ☒ a. kW-hour b. kW/hour c. kW d. All of the above
2. Rate at which energy is transferred or work is done. ☒ a. Energy ☒ b. Power c. Fuel d. All of the above	5. Unit of power a. kW-hour b. kW/hour ☒ c. kW d. All of the above
3. It is the power consumption ☒ a. Energy b. Power c. Fuel d. All of the above	6. It is an energy consumption. a. Energy ☒ b. Power c. Fuel d. All of the above
	7. kcal/hour is ____ ☒ a. Energy ☒ b. Power c. Heat d. None of the above

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8. Which of the following are the forms of energy? a. Kinetic energy b. Potential energy c. Heat energy d. Electrical energy ☒ e. All of the above	11. Power generated at the combustion chamber of an internal combustion engine. a. Drawbar horsepower ☒ b. Brake horsepower ☒ c. Indicated horsepower d. All of the above
9. Which are the sources of power that can be used for the farm? a. Human b. Draft animal c. Internal combustion engine ☒ d. Tractor ☒ e. All of the above f. Two of the above	12. Power that is available at the driveshaft of a machine. a. Indicated horsepower b. Drawbar horsepower ☒ c. Brake horsepower d. All of the above
10. Power required for pulling an implement or equipment. ☒ a. Indicated horsepower ☒ b. Drawbar horsepower c. Brake horsepower d. All of the above	13. Which of the following is the application of kinetic energy in producing power commonly used in agriculture? a. Windpump b. Hydropower ☒ c. Steam turbine ☒ d. All of the above e. None of the above

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14. Increasing the power supply to agriculture will result to ____.

- a. more tasks can be completed at the right time
- b. greater areas can be farmed
- c. greater quantities of crops can be produced
- ☒ d. All of the above
- e. None of the above

15. Energy-powered machinery like tractors, combines, and irrigation pumps can ____.

- a. significantly reduce manual labor
- b. shorten the time required for tasks like land preparation, planting, harvesting, and irrigation.
- c. ensures that critical operations, such as planting and harvesting, are performed within optimal timeframes
- ☒ d. maximize crop yields and minimize losses
- ☒ e. All of the above
- f. Two of the above

16. Is the energy an object possesses due to its motion.

- a. Potential
- ☒ b. Kinetic
- c. electrical
- d. All of the above

17. $\frac{1}{2} m v^2$ is the formula for

- a. Potential energy
- ☒ b. Kinetic energy
- c. Pressure energy
- d. None of the above

18. Kinetic energy is ____ related to the movement of an object.

- ☒ a. not
- ☒ b. directly
- c. indirectly
- d. One of the above

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19. Heavier objects have ____ kinetic energy at the same speed than lighter objects.

- a. less
- ☒ b. more
- c. no
- d. All of the above

20. The faster an object moves, the ____ its kinetic energy.

- ☒ a. lesser
- ☒ b. greater
- c. same
- d. None of the above

21. Energy associated with the movement of an object from one place to another.

- ☒ a. Translational
- b. Rotational
- c. Vibrational
- d. All of the above

22. Energy associated with the spinning or rotation of an object.

- a. Translational
- ☒ b. Rotational
- c. Vibrational
- d. All of the above

23. Energy associated with the movement of molecules or atoms.

- a. Translational
- b. Rotational
- ☒ c. Vibrational
- d. All of the above

24. The stored energy an object possesses due to its position, condition, or configuration.

- ☒ a. Kinetic
- ☒ b. Potential
- c. Pressure
- d. None of the above

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25. Potential energy is ____ energy in motion; it's energy that's held or stored within an object or system.

- a. not
- b. an
- c. Heat
- d. None of the above

26. The amount of potential energy an object has depends on its ____ relative to other objects or its internal configuration.

- a. position
- b. shape
- c. size
- d. All of the above

27. Potential energy can be converted into kinetic energy when the object's position or state ____.

- a. changes
- b. Is at rest
- c. Is absent
- d. All of the above

28. Which of the following is not a potential energy?

- a. Gravitational
- b. Elastic
- c. Chemical
- d. Electric
- e. All of the above
- f. None of the above

29. It essentially measures how quickly energy is being used or work is being performed.

- a. Power
- b. Force
- c. Work
- d. All of the above

30. Which of the following is the formula for power?

- a. Work / Time
- b. Energy / Time
- c. Force x Distance / Time
- d. All of the above
- e. None of the above

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31. An example of potential energy.

- a. 1 cubic meter of to be dropped from 10-m high
- b. 100 kg force to be moved in 20 meters
- c. 1 liter of diesel at ambient temperature
- d. All of the above
- e. Two of the above

32. Which of the following is higher?

- a. 1 Joule
- b. 1 N-m
- c. 0.737 ft-lb
- d. None of the above

33. One hundred BTU is equal to ____.

- a. 105.5 kJ
- b. 0.105 MJ
- c. 77817 ft-lb
- d. All of the above
- e. None of the above

34. Unit of energy

- a. Joule
- b. kcal
- c. British thermal unit
- d. kW-hr
- e. All of the above
- f. None of the above

35. Unit of power

- a. Watt
- b. kW
- c. MW
- d. Horsepower
- e. All of the above
- f. None of the above

36. Which of the following is not a unit of power?

- a. kcal/hr
- b. kg-m/sec
- c. BTU/hr
- d. None of the above
- e. All of the above

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37. Raising the temperature of water by heating it from 28° to 100°C is an example of ____.

- a. sensible heat
- b. latent heat of vaporization
- c. latent heat of condensation
- d. All of the above

38. The process of raising the temperature of food and water mixture to 100°C and then allowing it to simmer until the exothermic reaction in the food is completed.

- a. Steaming
- b. Boiling
- c. Baking
- d. None of the above

39. Moving air is an example of ____.

- a. potential energy
- b. kinetic energy
- c. sound energy
- d. None of the above

40. A form of energy associated with the position of the force field.

- a. Kinetic energy
- b. Potential energy
- c. Heat energy
- d. None of the above

41. The law that states, "there is loss in the system as the energy is transformed in the system".

- a. 1st law of Thermodynamics
- b. 2nd Law of Thermodynamics
- c. 3rd Law of Thermodynamics
- d. None of the above

42. The law that states, "energy can be created and destroyed".

- a. 1st law of Thermodynamics
- b. 2nd Law of Thermodynamics
- c. 3rd Law of Thermodynamics
- d. None of the above

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43. The kilo calories of heat used per unit time.

- a. Energy
- b. Power
- c. Work
- d. None of the above

44. Which of the following is not a unit of energy?

- a. MW-hour
- b. kW-hour
- c. Hp-hour
- d. None of the above
- e. All of the above

45. Which of the following is not a form of energy?

- a. Magnetic
- b. Nuclear
- c. Sound
- d. All of the above
- e. None of the above

46. A solenoid valve operates based on ____.

- a. sound energy
- b. kinetic energy
- c. magnetic energy
- d. None of the above

47. One joule per second is equal to ____.

- a. 1 watt
- b. 10 watts
- c. 1 kilowatt
- d. None of the above

48. One mega joule (MJ) is ____.

- a. 239 kcal
- b. 948 BTU
- c. 0.278 kW-hour
- d. All of the above

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49. A 6 MW power plant is operating for 24 hours per day for 4 months continuously, what is the energy delivered by the plant?

- a. 17,280 MW-hour
- b. 18,450 MW-hour
- c. 19,130 MW-hour
- d. None of the above

50. If a biomass power plant uses 7.5 tons per hour of biomass with a heating value of 3,000 kcal/kg, what is the input power of the plant?

- a. 25 MW
- b. 27 MW
- c. 30 MW
- d. None of the above

51. What is the expected power output of the plant above if the overall system thermal efficiency is 22%?

- a. 5.9MW
- b. 7.2 MW
- c. 9.1 MW
- d. None of the above

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52. Formula for human power.

- a. $0.1 \times \text{number of persons}$
- b. $0.2 \times \text{number of persons}$
- c. $0.3 \times \text{number of persons}$
- d. All of the above

53. Formula for Engine Indicated Horsepower.

- a. $P L A N n / (f \times 2)$
- b. $P L A N n / (f \times 1)$
- c. All of the above
- d. None of the above

54. Formula for the biomass power for thermal application.

- a. $FCR \times HVF \times 0.0012$
- b. $FCR \times HVF \times 0.0012 \times 0.6$
- c. $FCR \times HVF \times 0.0012 \times 0.2$
- d. All of the above

55. Formula for the biomass power for Electricity Generation

- a. $FCR \times HVF \times 0.0012$
- b. $FCR \times HVF \times 0.0012 \times 0.6$
- c. $FCR \times HVF \times 0.0012 \times 0.2$
- d. All of the above

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56. Formula for wind power output for pumping water.

- a. $\frac{1}{2} \rho a \times A r \times V^3$
- b. $\frac{1}{2} \rho a \times A r \times V^3 \times 0.2$
- c. $\frac{1}{2} \rho a \times A r \times V^3 \times 0.1$
- d. All of the above

57. Formula for wind power output for electricity generation.

- a. $\frac{1}{2} \rho a \times A r \times V^3$
- b. $\frac{1}{2} \rho a \times A r \times V^3 \times 0.2$
- c. $\frac{1}{2} \rho a \times A r \times V^3 \times 0.1$
- d. All of the above

58. Formula for Engine Brake Horsepower.

- a. $IHP \times \text{Mech Efficiency}$
- b. $IHP - FHP$
- c. All of the above
- d. None of the above

59. Formula for animal drawbar power

- a. $\text{Weight of Animal} \times \text{Speed of the animal}$
- b. $0.1 \times \text{Weight of the animal} \times \text{speed of the animal}$
- c. $0.2 \times \text{Weight of Animal} \times \text{Speed of the animal}$
- d. All of the above

60. Formula for hydro power in generating electricity

- a. $f Q H$
- b. $f Q H \epsilon t \epsilon g$
- c. $f Q H \epsilon t$
- d. All of the above

61. Formula for hydro power in generating mechanical power.

- a. $f Q H$
- b. $f Q H \epsilon t \epsilon g$
- c. $f Q H \epsilon t$
- d. All of the above

62. Formula for solar thermal power output.

- a. $c T_F A \cos \alpha \times \epsilon c$
- b. $c T_F A \cos \alpha$
- c. $\text{Power rating of module} \times \text{number of modules}$
- d. None of the above

63. kWe

- a. Unit use for expressing the electrical power
- b. Unit use for expressing the electrical power
- c. Unit use for expressing the electrical power
- d. None of the above

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